Vincent S. Pronel

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Education

Bachelor of Science in Computer Science **Expected 2027**
Kean University Union, NJ

- **Relevant Coursework**: Database Management Systems, Applied Statistics, Matrix & Linear Algebra, Analysis of Algorithms*, Software Engineering*, Object Oriented Analysis & Design **in progress, Fall 2026*
- **Leadership & Honors**: Web Developer, WiCyS (Women in Cyber Security) Kean Chapter E-Board | GMiS Research Posters Competition | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | Dean's List, Spring 2024 and Fall 2025

Technical Skills

- **Data & Databases**: SQL, MySQL, PostgreSQL, Supabase, Cloudflare D1, relational schema design, data marts, aggregation and normalization, correlation analysis
- **Visualization**: Google Charts, interactive dashboards, scatter plots and trendlines, correlation matrices, chart design
- **Languages**: SQL, PHP, Python, JavaScript, TypeScript, Java, HTML/CSS
- **Platforms & Tools**: MySQL Workbench, Apache/XAMPP, Next.js, React, Node.js, REST APIs, Cloudflare, Vercel, Git/GitHub, Linux

Research Projects

Java Execution Visualizer | *CRA UR2PhD-REU Fellowship* June 2026 – Present
Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, $8,000 funded, 400 hours *Kean University*

- Independently built a Java memory visualizer with 4 guided simulations and 26 execution states, representing stack frames, heap objects, arrays, aliasing, method calls, and reference changes line by line.
- Designed the **Supabase/PostgreSQL** data layer for an IRB-approved 2-condition study targeting 40 completions, with balanced assignment, idempotent event capture, hashed session tokens, an 11-event audit trail, and paired pre/post analytics.

Jobs vs. Wages | *NSF IUSE Scholars Academy* May 2026 – Present
NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, $1,400 funded, 10 weeks *Kean University*

- Built a population-weighted **MySQL data mart** from 33,000 records in the US ZIP Codes Database, writing the schema and aggregation logic for all 50 states, Washington, D.C., and Puerto Rico.
- Developed an interactive dashboard in PHP, JavaScript, and **Google Charts** with selectable variable pairs, live scatter plots with linear trendlines and R^2, a full correlation matrix, and a multi-state comparison view.
- Disproved the study's premise: unemployment explains only 14% of variation in household income ($r = -0.38$), while college attainment ($r = 0.79$) and poverty rate ($r = -0.74$) are more than twice as predictive; presented the findings at the 2026 GMiS conference in Albuquerque.

AI Debate Coach Evaluation | *CRA UR2PhD Research Program* Feb. 2026 – Apr. 2026
Undergraduate Researcher, Advisor: Dr. Yan Ma *Kean University*

- Prepared 120 OpenAI-compatible JSONL debate conversations comprising 960 messages across 75 multi-turn and 45 single-turn examples for a GPT-4o mini supervised fine-tuning pipeline.
- Built a 3-condition evaluation platform capturing transcripts, 24 Likert ratings per participant, comparative rankings, qualitative feedback, GPT-4o evaluation scores, and exports backed by Neon/PostgreSQL.

Professional Experience

Independent Web Developer Jan. 2026 – Present
Freelance Full-Stack Web Development *Sayreville, NJ*

- Independently delivered 5 production websites for 5 paying clients and built the WiCyS Kean Chapter site; design each **relational schema** and data layer and build admin tools used by approximately 2 staff members per business.
- Selected work: daysicecream.com (4,000+ monthly visitors) | trainingaio.com (booking platform, delivered in 2 weeks) | tedishairstudio.com | deafsharkcoffee.com | adzio.io | wicyskean.me.

Computer Science Tutor (Code Samurai) Jan. 2026 – Present
Kean University *Union, NJ*

- Selected by CS faculty to provide one-on-one Database Management Systems tutoring, coaching undergraduates through **relational schema design**, **SQL** queries, and step-by-step debugging.

Data Visualization & Analytics Intern July 2026
Group Summer Scholars Research Program, Kean University *Union, NJ*

- Lectured daily and mentored 5 high school students under *Dr. Ching-yu Huang*, teaching relational querying and interactive charting with SQL, MySQL, PHP, JavaScript, and Google Charts.
- Independently developed 8 hands-on labs supporting the faculty-led curriculum; all 5 students completed database-driven websites and presented individual research posters.